PROJECT DOCUMENTATION

Biker Gallery

Biker Rental Management System (BRMS)

A Web-Based Bike Rental Platform

for Kanchipuram, Tamil Nadu

Proprietors: Nithish & Padmasaran

Location: Biker Gallery, Kanchipuram, Tamil Nadu, India

Availability: 24/7

Year: 2026

Table of Contents

1. Introduction

Biker Gallery is a web-based Biker Rental Management System (BRMS) developed to simplify and digitise the operations of a bike rental business located in Kanchipuram, Tamil Nadu, India. The system is owned and operated by proprietors Nithish and Padmasaran. The primary objective of the BRMS is to allow customers to browse available bikes, register accounts, book rides, and complete payments online — all through a single, integrated web application.

Before the development of this system, the bike rental process was handled manually, which was time-consuming and prone to errors such as double bookings, inaccurate billing, and inadequate record maintenance. The BRMS replaces these manual processes with an automated platform that handles user registration, bike browsing, booking, payment processing, and administrative management.

The system is built using standard web technologies: HyperText Markup Language 5 (HTML5), Cascading Style Sheets 3 (CSS3), and JavaScript (JS) for the client-side interface, and PHP along with MySQL for the server-side logic and data storage. The application is designed to run on a local server environment using XAMPP (Apache + MySQL + PHP).

This documentation describes the complete technical and functional details of the Biker Gallery BRMS, including its modules, database structure, user interface design, and system diagrams.

2. Project Overview

2.1 Project Name and Purpose

The project is named Biker Gallery. Its purpose is to provide an end-to-end digital solution for bike rentals in Kanchipuram. The system enables customers to register, browse available bikes across multiple categories, select a preferred bike, make a booking by choosing a date and time slot, and pay a refundable deposit through multiple payment methods including Credit/Debit Card, Unified Payments Interface (UPI), Quick Response (QR) Code, and Net Banking.

2.2 Business Context

Biker Gallery is a local bike rental service based in Kanchipuram, Tamil Nadu. The proprietors offer premium, well-maintained bikes to customers at affordable hourly and daily rates, starting from Rs.18 per hour. The business operates around the clock and currently maintains a fleet of ten bikes across three categories: 125cc commuter and sports bikes, 200cc sport and naked bikes, and premium bikes above 200cc.

2.3 Scope of the System

The BRMS covers the following functional areas:

- Customer self-registration with identity verification using PAN Card and Driving License uploads with Optical Character Recognition (OCR) scanning.
- Secure login for both regular customers and administrators.
- Browsable and filterable bike catalogue (menu).
- Time-slot based booking system with real-time availability checking.
- Multi-method online payment with confirmation popups.
- Admin dashboard displaying booking statistics, revenue, and bike availability.
- Admin ability to mark bikes as returned and calculate refund amounts.
- Contact form for customer enquiries.

2.4 Technology Stack

Layer	Technology	Purpose
Frontend	HTML5, CSS3, JavaScript (ES6+)	User interface and client-side logic
Backend	PHP 8.x	Server-side processing and API endpoints
Database	MySQL (utf8mb4)	Persistent data storage
Server	Apache via XAMPP	Local web server environment
OCR Library	Tesseract.js v5	Document scanning during registration
Security	SHA-256 + bcrypt	Password hashing and verification

3. System Requirements

3.1 Hardware Requirements

Component	Minimum Specification
Processor	Intel Core i3 or equivalent (1.5 GHz or above)
RAM	4 GB (8 GB recommended)
Storage	500 MB free disk space for application files
Display	1024 x 768 resolution or higher
Internet	Required for OCR library (Tesseract.js CDN) and font loading

3.2 Software Requirements

Software	Version / Details
Operating System	Windows 10/11, macOS, or Linux
Web Server	Apache (included in XAMPP)
PHP Runtime	PHP 8.0 or above
Database Server	MySQL 5.7 or above (included in XAMPP)
Web Browser	Google Chrome (latest), Mozilla Firefox (latest), Microsoft Edge (latest), Safari (latest)

XAMPP	Version 8.x recommended
Code Editor (Dev)	Visual Studio Code, Notepad++, or any text editor

3.3 Supported Browsers

The application has been tested and confirmed to work correctly on the following browsers: Google Chrome (latest version), Mozilla Firefox (latest version), Microsoft Edge (latest version), Apple Safari (latest version), and mobile browsers including iOS Safari and Chrome for Android.

4. System Architecture

The Biker Gallery BRMS follows a traditional three-tier architecture pattern consisting of a Presentation Layer (client-side HTML/CSS/JS), an Application Layer (PHP server-side scripts), and a Data Layer (MySQL database). The following diagram illustrates the complete system architecture.

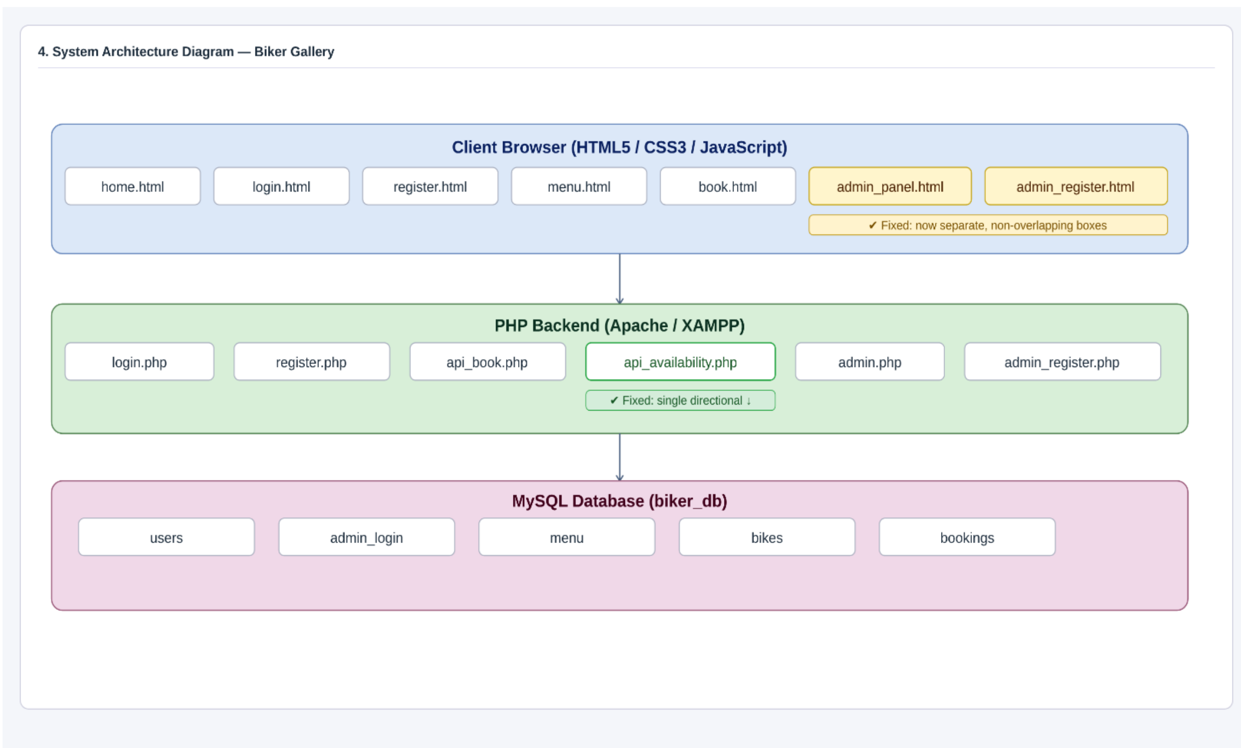

Figure 1: System Architecture Diagram

4.1 Presentation Layer

The presentation layer consists of all HTML pages that the user interacts with through the web browser. Each page is linked to dedicated CSS stylesheets for styling and JavaScript files for client-side interactivity. The pages include: home.html, login.html, register.html, menu.html, book.html, payment.html, contact.html, admin_panel.html, and admin_register.html.

4.2 Application Layer

The application layer is implemented in PHP and resides in the php/ subfolder. It processes all form submissions, performs server-side validation, queries the MySQL database, and returns responses (either HTTP redirects or JavaScript Object Notation (JSON) data for Application Programming Interface (API) endpoints). The PHP files include: login.php, register.php, admin_register.php, api_book.php, api_availability.php, admin.php, and db.php.

4.3 Data Layer

The data layer consists of a MySQL database named "biker" containing five tables: users, admin_login, menu, bikes, and bookings. All tables use the utf8mb4 character set with unicode collation to support all characters including Indian Rupee symbols and Unicode text.

5. Module Description

The BRMS consists of the following major modules, each serving a specific functional role within the system.

5.1 User Registration Module

The registration module (register.html and php/register.php) enables new customers to create an account on the Biker Gallery platform. The module collects personal information including username, full name, date of birth, email address, phone number, and password. Additionally, the module requires the upload of two identity documents: a PAN Card and a Driving License. Both documents are scanned using the Tesseract.js OCR library (js/scanner.js) to extract and display text for verification purposes.

Client-side validation is applied to all fields using the register.js script. Validation rules include: username must be a minimum of four characters; full name must consist of letters and spaces only (three to thirty-five characters); the applicant must be at least eighteen years of age; email must follow a valid format; phone number must be exactly ten digits; and password must be between six and twenty characters. A password show/hide toggle is provided for convenience.

On the server side (register.php), the system verifies that the submitted email and username are not already registered in the users table. The password arrives as a SHA-256 hash (computed client-side by register.js), and the server applies bcrypt hashing before storing it in the database, resulting in a double-hashed password for enhanced security.

5.2 User Login Module

The login module (login.html and php/login.php) provides authentication for two types of users: regular customers and administrators. The login form collects a username and password. The JavaScript file js/login.js handles client-side validation and SHA-256 hashes the password before submission.

On the server side (login.php), the system reads the hidden field login_authentication to determine the authentication route. A value of "1" triggers a lookup in the users table for regular customer login, redirecting to home.html on success. A value of "admin" triggers a lookup in the admin_login table, redirecting to admin.php on success. A floating button styled as an administrator icon is positioned at the top-right corner of the login page to provide administrators with a distinct login entry point.

5.3 Home Page Module

The home page (home.html) is the landing page displayed after a successful customer login. It presents the Biker Gallery brand with a hero section containing the tagline "Ride Smart, Pay Less," key statistics (number of bikes available, starting price, and operating hours), and feature highlight cards. The page provides quick navigation links to browse bikes, book a ride, and contact the proprietors. The footer displays the copyright year, proprietors' names, and location.

5.4 Bike Menu Module

The menu module (menu.html and js/menu.js) presents the complete fleet of ten bikes available for rental. Bikes are grouped into three categories: 125cc, 200cc, and Premium. A filter row allows customers to view all bikes or filter by category. Each bike card displays the bike image, name, engine specification, mileage, gearbox type, bike type, hourly rate, daily rate, and an availability status chip. A "Book Now" button on each card navigates directly to the booking page with the selected bike's name and rate pre-populated via URL query parameters.

The complete bike fleet is as follows:

Bike Name	Category	Engine	Mileage	Gearbox	Type	Rate/hr	Rate/day
Honda SP 125	125cc	125cc	65 kmpl	Manual	Commuter	Rs.20	Rs.480
Hero Xtreme 125R	125cc	125cc	60 kmpl	Manual	Sports	Rs.22	Rs.520

Bajaj Pulsar NS125	125cc	125cc	55 kmpl	Manual	Sports	Rs.22	Rs.520
Honda Activa 125	125cc	125cc	60 kmpl	Automatic	Scooter	Rs.18	Rs.440
Bajaj Pulsar NS200	200cc	200cc	40 kmpl	Manual	Sports	Rs.35	Rs.800
TVS Apache RTR 200 4V	200cc	197cc	35 kmpl	Manual	Sports	Rs.38	Rs.880
Yamaha MT-15	200cc	155cc	43 kmpl	Manual	Naked	Rs.40	Rs.920
Hero XF3R	Premium	210cc	30 kmpl	Manual	Sports	Rs.60	Rs.1400
Honda CBR300R	Premium	286cc	28 kmpl	Manual	Sport	Rs.75	Rs.1700
Benelli TNT 300	Premium	300cc	25 kmpl	Manual	Naked	Rs.80	Rs.1800

5.5 Booking Module

The booking module (book.html and js/book.js) enables customers to reserve a bike for a specific date and time slot. The booking form displays the selected bike's name, availability status, and hourly rate at the top. The customer fills in their full name, phone number, and pickup date. Once a date is selected, the system calls the availability API (php/api_availability.php via an asynchronous GET request) to retrieve blocked time slots for the chosen bike and date. Available time slots from 8:00 AM to 10:00 PM are displayed as interactive buttons.

A booking summary panel shows the selected date, pickup time, a refundable deposit amount of Rs.2,500, and the total payable amount. The deposit is always Rs.2,500, regardless of the hourly rate or duration, as the actual rental duration is determined upon the bike's return by the admin. Clicking "Proceed to Payment" validates all fields and redirects to payment.html with the booking details stored in the browser's session storage.

5.6 Payment Module

The payment module (payment.html and js/payment.js) handles the collection of the refundable deposit from customers. The page displays an Order Summary card showing the bike name, customer name, booking date and time, base subtotal, Goods and Services Tax (GST) at 18%, subtotal including GST, the refundable deposit amount, and the total payable amount.

Four payment method tabs are provided: Card (Credit/Debit), UPI, QR Code, and Net Banking. The Card panel collects the cardholder name (validated to accept letters and spaces only using the fmtCardName function), card number (formatted as groups of four digits), and expiry date. The UPI panel allows selection of a UPI application. The QR Code panel displays a static QR code for scanning. The Net Banking panel provides bank selection.

Before processing, a JavaScript confirm() popup is displayed showing the selected payment method and total amount. A separate confirmation popup is shown when the customer attempts to navigate back to the home page, warning that the booking will be cancelled. Upon successful payment, a success overlay is displayed. On failure, a failure overlay with a retry option is shown.

5.7 Contact Module

The contact module (contact.html and js/contact.js) provides contact information for the Biker Gallery proprietors and includes a simple enquiry form. The page displays Proprietor 1 (Nithish), Proprietor 2 (Padmasaran), phone/WhatsApp number, working hours (24/7), location (Kanchipuram, Tamil Nadu), and email address. The enquiry form accepts the customer's name, phone number, and message. Validation ensures the name and phone fields are properly filled before submission.

5.8 Admin Panel Module

The admin panel module (admin_panel.html rendered by php/admin.php) is accessible only to authenticated administrators. The dashboard displays four statistical summary cards: total bikes currently booked, total bikes available, total revenue excluding GST (from returned bookings), and total amount charged including GST. Two data tables are shown: the Active Bookings table listing all currently rented bikes with customer details, pickup date, pickup time, deposit amount, status chip, and a "Mark Available" action button; and the Booking History table showing all returned bookings with actual hours used, amount charged, and refund calculated.

When the admin clicks "Mark Available" for an active booking, a modal dialog prompts for the number of actual hours the bike was used. The system then calls the api_availability.php endpoint via a POST request with action "set_available," calculates the actual rental charge (rate per hour multiplied by actual hours plus 18% GST), and computes the refund amount as the deposit minus the total charge. The booking is updated to "returned" status in the database.

5.9 Admin Registration Module

The admin registration module (admin_register.html, php/admin_register.php, css/admin_register.css, and js/admin_register.js) allows the creation of new administrator accounts. The form collects the admin username, full name, email address, phone number, and password. All fields undergo live and on-blur validation using the admin_register.js script. The password is SHA-256 hashed on the client side before transmission, and the server applies bcrypt hashing for storage in the admin_login table.

6. User Interface (UI) Form Design

This section presents the user interface screens of the Biker Gallery BRMS. All screenshots shown below are taken directly from the uploaded project files.

6.1 Home Page

The Home Page serves as the main landing page after login. It displays the Biker Gallery brand name, a navigation bar, a hero section with the tagline and call-to-action buttons, statistical highlights (number of bikes, starting price, availability, and location), and feature cards. The footer contains the copyright notice and proprietors' names.

6.2 Login Page

The Login Page provides a login form with fields for username and password. A password show/hide toggle button is positioned inside the password field. A floating administrator icon button in the top-right corner triggers the admin login flow. Error messages are displayed inline below each field. A registration link at the bottom directs new users to the registration page.

6.3 Registration Page

The Registration Page collects personal information from new customers. The form includes fields for username, full name, date of birth, email address, phone number, password (with show/hide toggle), and confirm password. Below the form fields, two document upload zones accept JPG and PNG files for PAN Card and Driving License. The OCR scanning process is initiated automatically on upload, and extracted text is displayed in result cards below each upload zone. A "Complete Registration" button submits the form after all validations pass.

6.4 Bike Menu Page

The Menu Page presents the complete bike fleet in a card-based grid layout. At the top, a filter row with category buttons (All Bikes, 125cc, 200cc, Premium) allows customers to filter the displayed bikes. Each bike card includes a photograph of the bike, an availability chip, a category chip, the bike name, specification details in a grid format (engine, mileage, gearbox, type), the hourly and daily pricing, and a "Book Now" button. The following figures show the bikes available in each category.

Figure 2: Honda SP 125 — 125cc Commuter — Rs.20/hr

Figure 3: Hero Xtreme 125R — 125cc Sports — Rs.22/hr

Figure 4: Bajaj Pulsar NS125 — 125cc Sports — Rs.22/hr

Figure 5: Honda Activa 125 — 125cc Scooter — Rs.18/hr

Figure 6: Bajaj Pulsar NS200 — 200cc Sports — Rs.35/hr

Figure 7: TVS Apache RTR 200 4V — 200cc Sports — Rs.38/hr

Figure 8: Yamaha MT-15 — 200cc Naked — Rs.40/hr

Figure 9: Hero XF3R — Premium 210cc Sports — Rs.60/hr

Figure 10: Honda CBR300R — Premium 286cc Sport — Rs.75/hr

Figure 11: Benelli TNT 300 — Premium 300cc Naked — Rs.80/hr

6.5 Booking Page

The Booking Page displays the selected bike's name and rate at the top with a "change" link back to the menu. The form collects the customer's full name, phone number, and pickup date. After a date is selected, available time slots from 8:00 AM to 10:00 PM are rendered as clickable buttons colour-coded as available (green), booked (red), or selected (indigo). A booking summary panel below the slots shows the selected date, pickup time, refundable deposit (Rs.2,500), and total payable. The "Proceed to Payment" button validates the form and redirects to the payment page.

6.6 Payment Page

The Payment Page is divided into two sections. The left section (Order Summary) presents a detailed breakdown including bike name, customer name, booking date, pickup time, rental duration, base subtotal, GST (18%), subtotal including GST, refundable deposit, and the final total payable. The right section (Payment Card) offers four payment method tabs: Card, UPI, QR Code, and Net Banking. The Card tab includes fields for cardholder name, card number, and expiry date. A confirmation popup is displayed before payment processing is initiated.

6.7 Contact Page

The Contact Page presents the proprietors' contact details in an information card grid, including names, phone number, working hours, location, and email. A message form below the contact details allows customers to submit enquiries with their name, phone number, and a message. A success notification is displayed upon successful submission.

6.8 Admin Panel

The Admin Panel (accessible to administrators only) features a header with the dashboard title and a "Register New Admin" icon button. Four summary cards show key metrics: bikes booked, bikes available, revenue excluding GST, and total charged including GST. The Active Bookings table lists all currently rented bikes with columns for serial number, customer details, bike name, pickup date, pickup time, deposit amount, booking status, and an action button. The Booking History table shows completed rentals with actual hours used, charges applied, and refund amounts calculated.

6.9 Admin Registration Page

The Admin Registration Page provides a form for registering new admin accounts. The indigo-themed form collects username, full name, email address, phone number, password (with show/hide toggle), and confirm password. All fields undergo real-time validation. The form is submitted to admin_register.php, which applies bcrypt hashing to the SHA-256 pre-hashed password before storing the credentials.

7. Database Design

The Biker Gallery BRMS uses a single MySQL database named "biker" (configured in php/db.php with the database name "bikerr"). The database contains five tables, each described in detail below. The character set is utf8mb4 with unicode collation to support full Unicode character encoding.

7.1 Table: users

The users table stores registered customer account information.

Column Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT, NOT NULL	Unique customer identifier
username	VARCHAR(50)	NOT NULL, UNIQUE	Customer login username
full_name	VARCHAR(100)	NOT NULL	Customer full name
dob	DATE	NOT NULL	Customer date of birth
email	VARCHAR(100)	NOT NULL, UNIQUE	Customer email address
phone	VARCHAR(10)	NOT NULL	Customer phone number (10 digits)
password	VARCHAR(255)	NOT NULL	bcrypt(SHA-256 hash) of password
created_at	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Account creation timestamp

7.2 Table: admin_login

The admin_login table stores administrator account credentials.

Column Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT, NOT NULL	Unique admin identifier
full_name	VARCHAR(100)	NOT NULL	Admin full name
username	VARCHAR(50)	NOT NULL, UNIQUE	Admin login username
email	VARCHAR(100)	NOT NULL, UNIQUE	Admin email address
phone	VARCHAR(20)	NOT NULL	Admin phone number
password	VARCHAR(255)	NOT NULL	bcrypt(SHA-256 hash) of password
created_at	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Account creation timestamp

7.3 Table: menu

The menu table stores the names of all bikes available in the rental catalogue. It is used to validate bike names during the booking process.

Column Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT, NOT NULL	Unique menu entry identifier
bike_name	VARCHAR(100)	NOT NULL, UNIQUE	Name of the bike as listed in the menu

7.4 Table: bikes

The bikes table stores detailed specifications and pricing for each bike in the fleet.

Column Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique bike record identifier
bike_name	VARCHAR(100)	NOT NULL, UNIQUE	Name of the bike
category	ENUM('125cc','200cc','premium')	NOT NULL	Bike category for filtering
engine	VARCHAR(20)	NOT NULL	Engine displacement
mileage	VARCHAR(20)	NOT NULL	Fuel efficiency in kmpl
gearbox	VARCHAR(20)	NOT NULL	Transmission type (Manual/Automatic)
type	VARCHAR(30)	NOT NULL	Bike type (Sports, Commuter, Naked, etc.)
price_hr	INT	NOT NULL	Rental price per hour in Rs.
price_day	INT	NOT NULL	Rental price per day in Rs.
img_path	VARCHAR(100)	NOT NULL	Relative path to the bike image file

7.5 Table: bookings

The bookings table is the central operational table of the BRMS. It stores all booking records including deposit details, actual usage hours, and refund calculations.

Column Name	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT, NOT NULL	Unique booking identifier
bike_name	VARCHAR(100)	NOT NULL, INDEX	Name of the booked bike
customer_name	VARCHAR(100)	NOT NULL	Customer full name
phone	VARCHAR(10)	NOT NULL	Customer phone number
pickup_date	DATE	NOT NULL, INDEX	Date of bike pickup
pickup_time	TIME	NOT NULL, INDEX	Time of bike pickup
hours	INT	NOT NULL	Booked hours (stored as 0 for open-ended)
rate	INT	NOT NULL	Hourly rental rate at time of booking
gst	INT	NOT NULL	GST amount (18%) — set to 0 at booking
total	INT	NOT NULL	Total amount paid (deposit at booking time)
deposit	INT	NOT NULL, DEFAULT 2500	Refundable security deposit amount
actual_hours	INT	NULL	Actual hours used (set by admin on return)
refund_amount	INT	NULL	Deposit minus actual charge (set on return)
status	ENUM('active','returned')	NOT NULL, DEFAULT 'active', INDEX	Booking status
returned_at	TIMESTAMP	NULL	Timestamp when bike was returned
booked_at	TIMESTAMP	NOT NULL, DEFAULT CURRENT_TIMESTAMP	Timestamp when booking was created

7.6 Deposit and Refund Calculation Logic

At the time of booking, the customer pays only the refundable deposit of Rs.2,500. The actual rental charge is computed when the admin marks the bike as returned. The refund calculation formula is:

Actual Base Charge = Rate per Hour x Actual Hours Used

GST Amount = Actual Base Charge x 0.18

Total Actual Charge = Actual Base Charge + GST Amount

Refund Amount = Deposit (Rs.2,500) - Total Actual Charge

A positive refund amount means the customer receives money back. A negative refund amount means the customer owes the additional amount to the proprietors.

8. Flow Diagram

The following diagram illustrates the complete operational flow of the Biker Gallery BRMS from the point where a user visits the website to the completion of a booking and the subsequent admin action of marking the bike as returned.

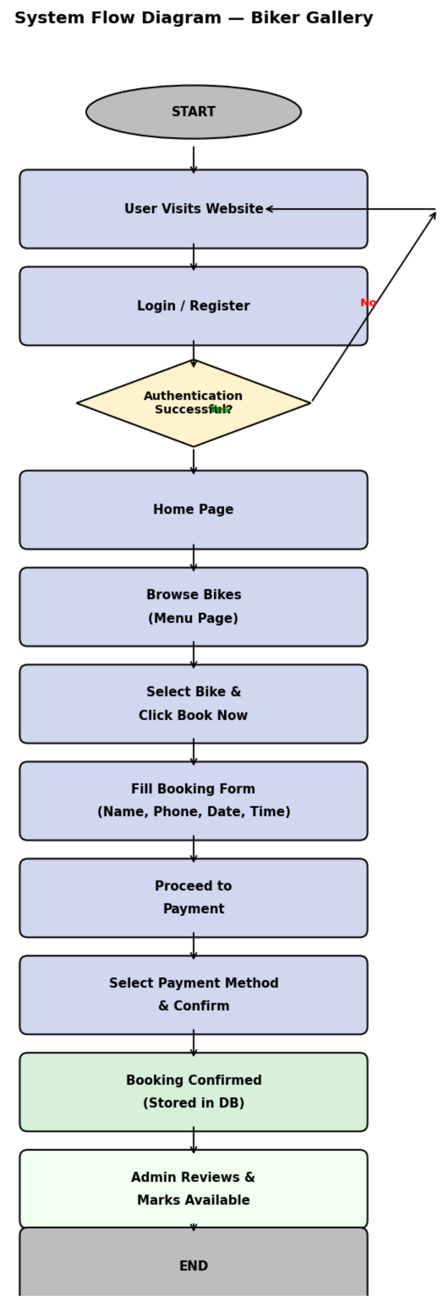

Figure 12: System Flow Diagram

As shown in the flow diagram, the process begins when a user visits the website. The user is presented with the login/register option. If authentication fails, the user is redirected back to the login page. Upon successful authentication, the user reaches the home page and can browse the bike menu. After selecting a bike, the user fills in the booking form, selects an available time slot, and proceeds to payment. Once payment is confirmed, the booking is stored in the database with a status of "active." The admin subsequently reviews active bookings and marks the bike as available upon physical return, triggering the refund calculation.

9. Use Case Diagram

The Use Case Diagram below defines the interactions between the two primary actors of the system — the User (customer) and the Admin — and the functional use cases supported by the Biker Gallery BRMS.

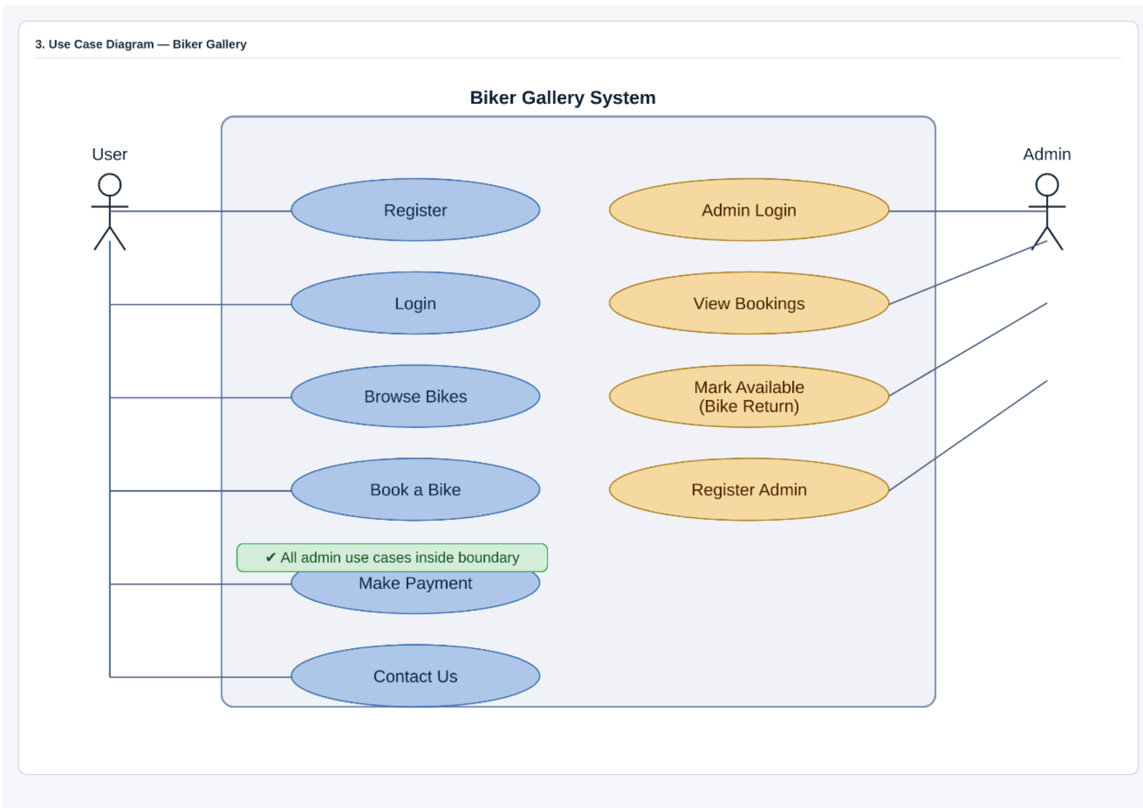

Figure 13: Use Case Diagram

9.1 User (Customer) Use Cases

- Register: Create a new customer account by submitting personal details and identity documents.
- Login: Authenticate using a username and password to access the system.

- Browse Bikes: View the full bike catalogue and filter bikes by category.
- Book a Bike: Select a bike, choose a date and time slot, and confirm a booking.
- Make Payment: Pay the refundable deposit using Card, UPI, QR Code, or Net Banking.
- Contact Us: Submit an enquiry message to the proprietors.

9.2 Admin Use Cases

- Admin Login: Authenticate via the admin login route to access the admin panel.
- View Bookings: View all active and historical bookings in the dashboard.
- Mark Available (Bike Return): Record actual hours, compute charges and refund, and update booking status to "returned."
- Register Admin: Create new administrator accounts via the admin registration form.

10. Class Diagram

The Class Diagram below represents the object-oriented structure of the Biker Gallery BRMS. It shows the primary classes, their attributes, methods, and the relationships between them.

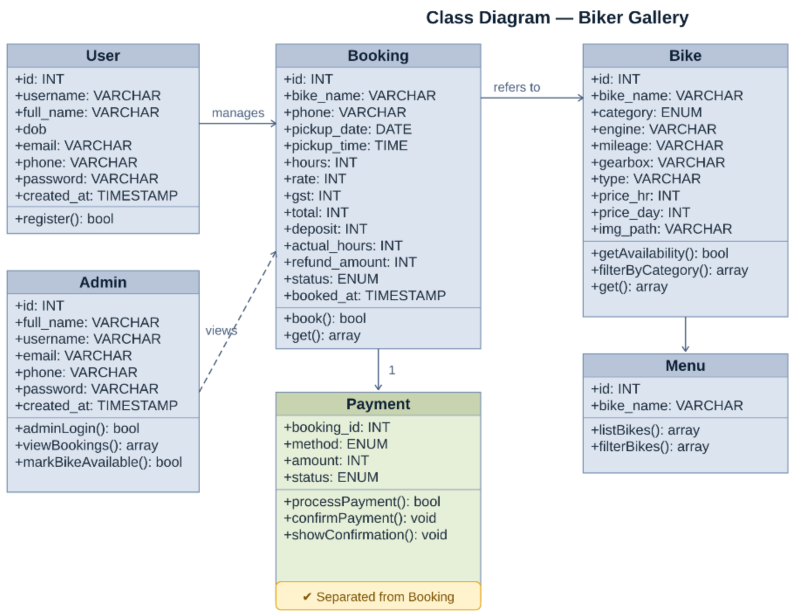

Figure 14: Class Diagram

The User class represents a registered customer with attributes corresponding to the users database table. Its methods include register(), login(), and logout(). The Booking class encapsulates all booking-related data and provides methods for creating bookings, marking bikes as returned, and calculating refund amounts. The Bike class stores all specifications for a rental bike and provides methods to check availability and filter bikes by category. The Admin class represents an administrator with credentials and methods to log in, view bookings, and mark bikes available. The Payment class manages payment method details and processing. The Menu class provides a catalogue listing of available bikes with filtering capability.

11. Sequence Diagram

The Sequence Diagram below illustrates the chronological message flow between system components during a complete bike booking transaction, from the initial page load through to the admin marking the bike as returned.

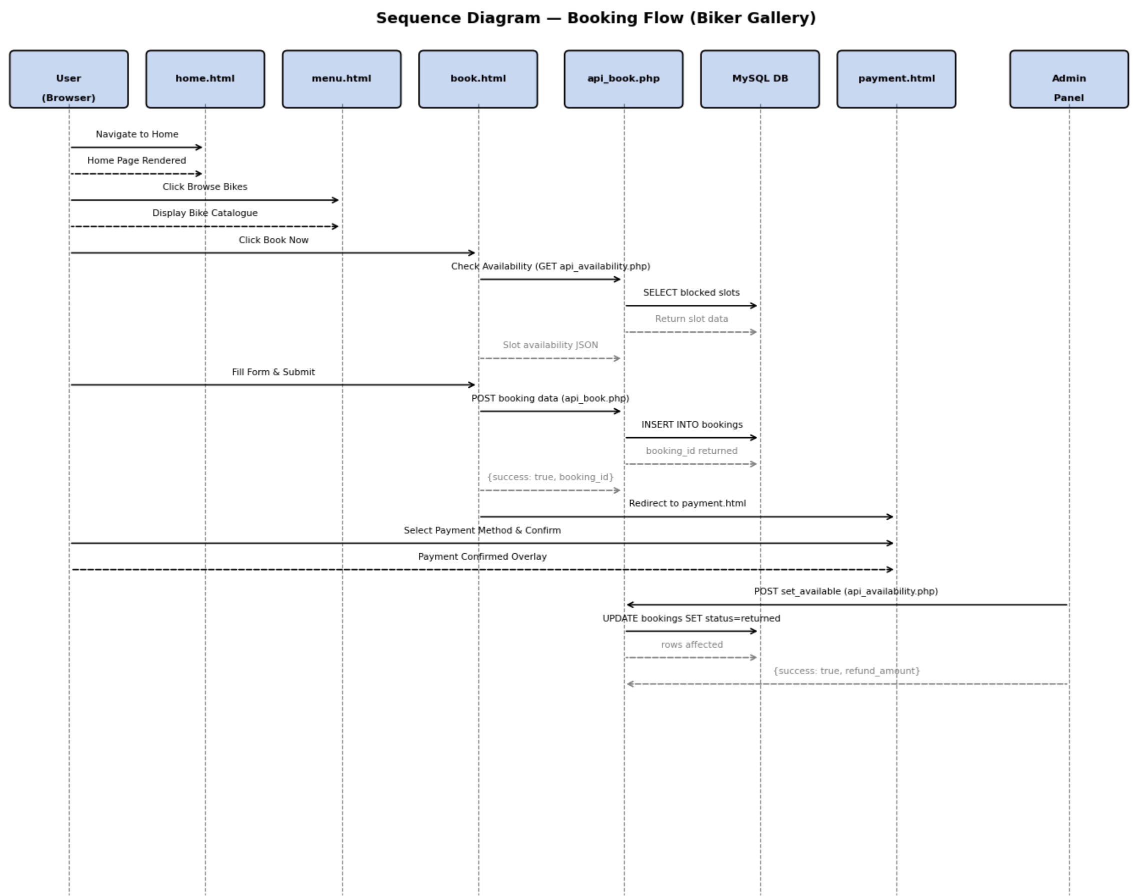

Figure 15: Sequence Diagram — Booking Flow

The sequence begins with the User navigating to the home page, which returns the rendered HTML. The user then navigates to the menu page to browse bikes and selects a bike to proceed to the booking page. The booking page calls the availability API (api_availability.php via a GET request) to retrieve blocked slots for the selected bike and date. The API queries the MySQL database and returns the slot data as a JSON response. The user fills in the booking form and submits it. The booking page sends a POST request to api_book.php, which inserts the new booking record into the MySQL database and returns a JSON response with the booking identifier. The booking page then redirects the user to payment.html. The user selects a payment method, confirms the payment popup, and the payment success overlay is displayed. Subsequently, the Admin accesses the admin panel, selects an active booking, and sends a POST request to api_availability.php with the action "set_available" and the actual hours used. The API updates the booking status in the database to "returned" and returns the calculated refund amount to the admin panel.

12. Entity-Relationship (ER) Diagram

The Entity-Relationship (ER) Diagram below provides a visual representation of the database schema for the Biker Gallery BRMS, showing all five tables, their attributes, primary keys (PK) and foreign key (FK) relationships, and the cardinality between entities.

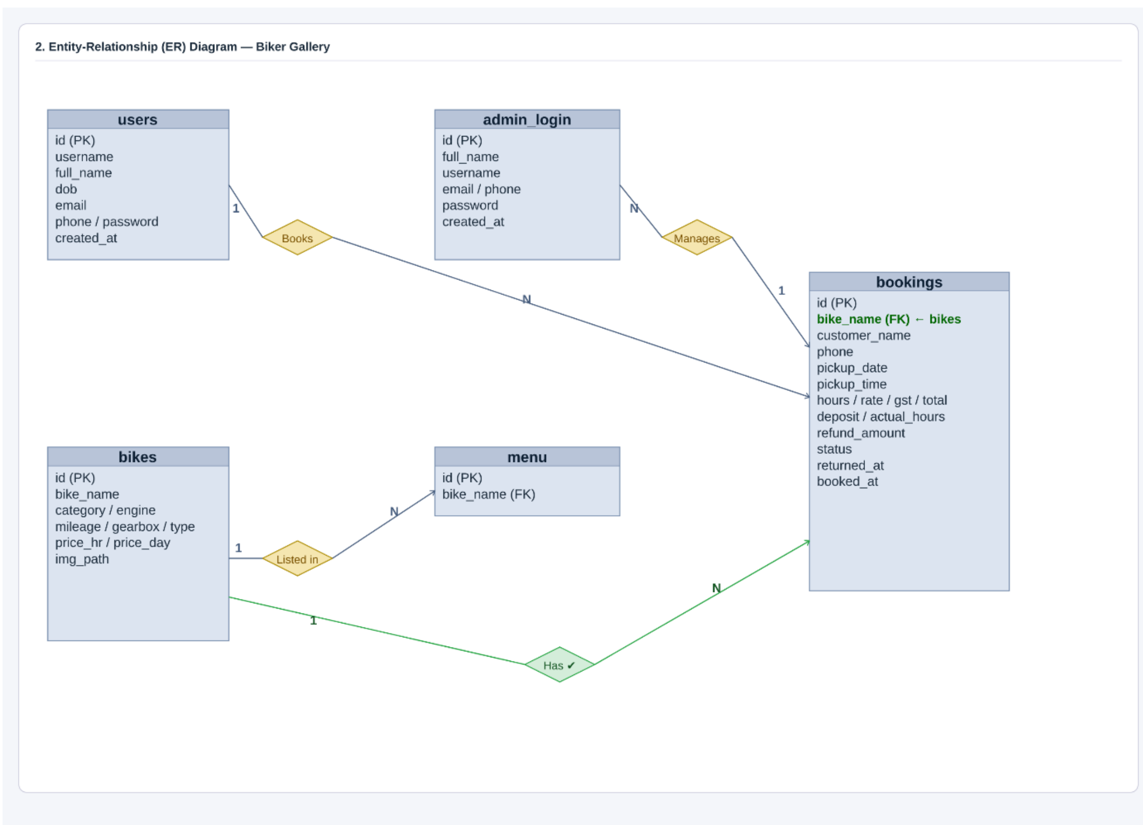

Figure 16: Entity-Relationship (ER) Diagram

The key relationships in the BRMS database are as follows:

- Users and Bookings: A single user (customer) can make multiple bookings. This is a one-to-many relationship from users to bookings, linked through the customer's name and phone details stored in each booking record.
- Bikes and Bookings: A single bike can be associated with multiple booking records over time. The bike_name field in the bookings table references the bike_name in the menu and bikes tables. At any given time, a bike can have only one active booking.
- Menu and Bikes: The menu table and bikes table both maintain bike names. The menu table serves as the reference for booking validation (api_book.php checks against the menu table to confirm the bike exists), while the bikes table stores full specifications used for display on the menu page.

The admin_login table is independent from the users table, maintaining a separate set of administrator credentials with no relational links to the operational booking data.

13. Security Implementation

The Biker Gallery BRMS implements several security measures to protect user data and prevent common web vulnerabilities.

13.1 Password Security

A two-phase password hashing approach is employed. On the client side, the user's plaintext password is hashed using the SHA-256 algorithm before it is transmitted to the server (implemented in js/login.js, js/register.js, and js/admin_register.js). This ensures that the plaintext password is never sent over the network. On the server side, the received SHA-256 hash is further hashed using PHP's password_hash() function with the PASSWORD_BCRYPT algorithm before being stored in the database. Authentication is performed using password_verify() to compare the submitted SHA-256 hash against the stored bcrypt hash.

13.2 SQL Injection Prevention

All database interactions across the PHP backend files use prepared statements with parameterised queries (MySQLi's prepare() and bind_param() methods). This approach ensures that user-supplied input is never directly concatenated into SQL queries, effectively preventing SQL injection attacks.

13.3 Input Validation

Both client-side and server-side input validation are implemented. Client-side validation (JavaScript) provides immediate user feedback for field-level errors. Server-side validation (PHP) re-validates all inputs upon receipt, regardless of client-side validation, to prevent manipulation of POST requests. Specific validation rules include phone number format checks, email format checks, date format checks using regular expressions, and cardholder name character restriction (letters and spaces only).

13.4 Session Management

PHP sessions are used to track authenticated users and administrators. The session is initiated with session_start() at the beginning of each PHP script. Session variables (user_id, username, logged_in for customers; admin_id, admin_username, admin_logged_in for administrators) are set upon successful login. The admin panel checks for the presence of admin session variables before rendering the dashboard content.

13.5 Cross-Origin Resource Sharing (CORS)

The API endpoints (api_book.php and api_availability.php) include CORS headers (Access-Control-Allow-Origin: *) to permit requests from the client-side JavaScript. In a production deployment, this header should be restricted to the specific domain of the application.

13.6 Document Verification

During customer registration, identity document verification is enforced by requiring the upload of both a PAN Card and a Driving License image. The Tesseract.js OCR library scans the uploaded images and extracts text for display, allowing the proprietors (or the system) to confirm document authenticity before fully activating an account.

14. Testing

14.1 Browser Compatibility Testing

The Biker Gallery BRMS has been tested across the following web browsers and confirmed to function correctly: Google Chrome (latest version), Mozilla Firefox (latest version), Microsoft Edge (latest version), Apple Safari (latest version), iOS Safari (mobile), and Google Chrome for Android (mobile). All UI components, JavaScript interactions, form validations, and API calls operate consistently across these environments.

14.2 Module-Level Testing Checklist

Module	Test Case	Expected Result	Status
Registration	Submit form with all valid fields	Account created; redirect to login	Pass
Registration	Submit with duplicate username	Error: Username already taken	Pass
Registration	Submit with duplicate email	Error: Email already registered	Pass
Registration	Upload PAN Card image	OCR scans and displays extracted text	Pass

Login	Login with valid user credentials	Session created; redirect to home.html	Pass
Login	Login with invalid password	Error message displayed; no session created	Pass
Admin Login	Login via admin button with valid credentials	Admin session created; redirect to admin.php	Pass
Menu	Click category filter buttons	Bikes filtered correctly by category	Pass
Menu	Click Book Now on a bike	Redirect to book.html with bike name and rate in URL	Pass
Booking	Select a date with available slots	Time slot grid renders correctly	Pass
Booking	Select a booked time slot	Slot shown as blocked; cannot be selected	Pass
Booking	Click Proceed to Payment without filling fields	Validation errors displayed for empty fields	Pass
Payment	Enter letters in cardholder name field	Only letters and spaces accepted; numbers rejected	Pass
Payment	Click Pay button	Confirmation popup appears with method and amount	Pass
Payment	Click Back to Home from payment page	Warning popup about booking cancellation shown	Pass
Payment	Complete payment with valid card details	Success overlay displayed	Pass
Admin Panel	Click Mark Available for active booking	Modal dialog for actual hours input shown	Pass
Admin Panel	Submit actual hours in Mark Available modal	Booking updated to returned; refund calculated and displayed	Pass
Admin Reg	Register new admin with valid data	Admin account created in admin_login table	Pass
Contact	Submit enquiry form with valid data	Success message displayed	Pass

14.3 Payment Module Specific Testing

The payment module includes specific test cases for the newly implemented features. The cardholder name field is validated in real time using the fmtCardName() function, which strips all non-alphabetic characters (digits and special characters) from the input as the user types. The processPayment() function performs an additional regular expression check (/^[a-zA-Zs]+$/) before submitting the payment. Both the Pay button confirmation popup and the Back to Home confirmation popup have been tested across all supported browsers using the native JavaScript confirm() dialog.

15. Conclusion

The Biker Gallery Biker Rental Management System (BRMS) is a complete, functional web application that successfully digitises the end-to-end operations of a bike rental business in Kanchipuram, Tamil Nadu. The system provides a seamless experience for customers, allowing them to register with identity document verification, browse a categorised fleet of ten bikes, book rides using a real-time time-slot system, and pay securely through multiple payment methods.

The admin panel equips the proprietors, Nithish and Padmasaran, with the tools needed to monitor booking activity, track revenue, manage deposit refunds, and maintain the availability status of their bike fleet — all from a single dashboard. The separation of the admin interface from the customer interface, combined with role-based session management, ensures that administrative functions remain secure and inaccessible to regular users.

The codebase follows sound software engineering practices including separation of concerns (HTML, CSS, and JavaScript maintained in separate files), server-side prepared statements for SQL injection prevention, double-phase password hashing (SHA-256 on the client side and bcrypt on the server side), and real-time client-side validation to provide immediate user feedback.

The BRMS is designed to run on a standard XAMPP environment, making it accessible for local deployment without the need for cloud infrastructure. The system is compatible with all major modern web browsers and mobile devices. Future enhancements could include email and SMS notifications for booking confirmations, integration with a live payment gateway for actual transaction processing, GPS-based bike tracking, and a mobile application interface for customers.

In summary, the Biker Gallery BRMS provides a robust, secure, and user-friendly platform that significantly improves the operational efficiency of the bike rental business while delivering a modern and convenient rental experience to customers in Kanchipuram and the surrounding region.

— End of Documentation —